1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

In men with BRCA2-deficient prostate cancer treated with Olaparib, therapy-induced DNA damage activates immune signaling that increases intratumoral TIL density as measured by imaging-based readouts. The magnitude of post-treatment TIL density (or change from baseline) will positively correlate with overall survival, making post-treatment TIL density a predictive biomarker of Olaparib benefit in BRCA2-mutant disease.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	BRCA2
Primary endpoint	OS

2. Why This Gene — Not Others

Primary hypothesis gene(s)

BRCA2: Selected as a core mechanistic driver from Tier 1 knowledge graph analysis.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	EDIH	Supporting modifier from knowledge graph
Secondary	EDIP	Supporting modifier from knowledge graph
Secondary	insulin_response_markers	Supporting modifier from knowledge graph

Secondary	metabolite_biomarkers	Supporting modifier from knowledge graph
Secondary	f/tPSA	Supporting modifier from knowledge graph
Secondary	CSF1	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: and one imaging feature (TIL density) to maintain testable, mechanistically grounded predictions; exploratory modifiers (EDIH, EDIP, insulin-response markers, metabolite biomarkers, f/tPSA, CSF

Prediction 2: can be evaluated in secondary analyses without altering the primary claim.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- EDIH: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- EDIP: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.